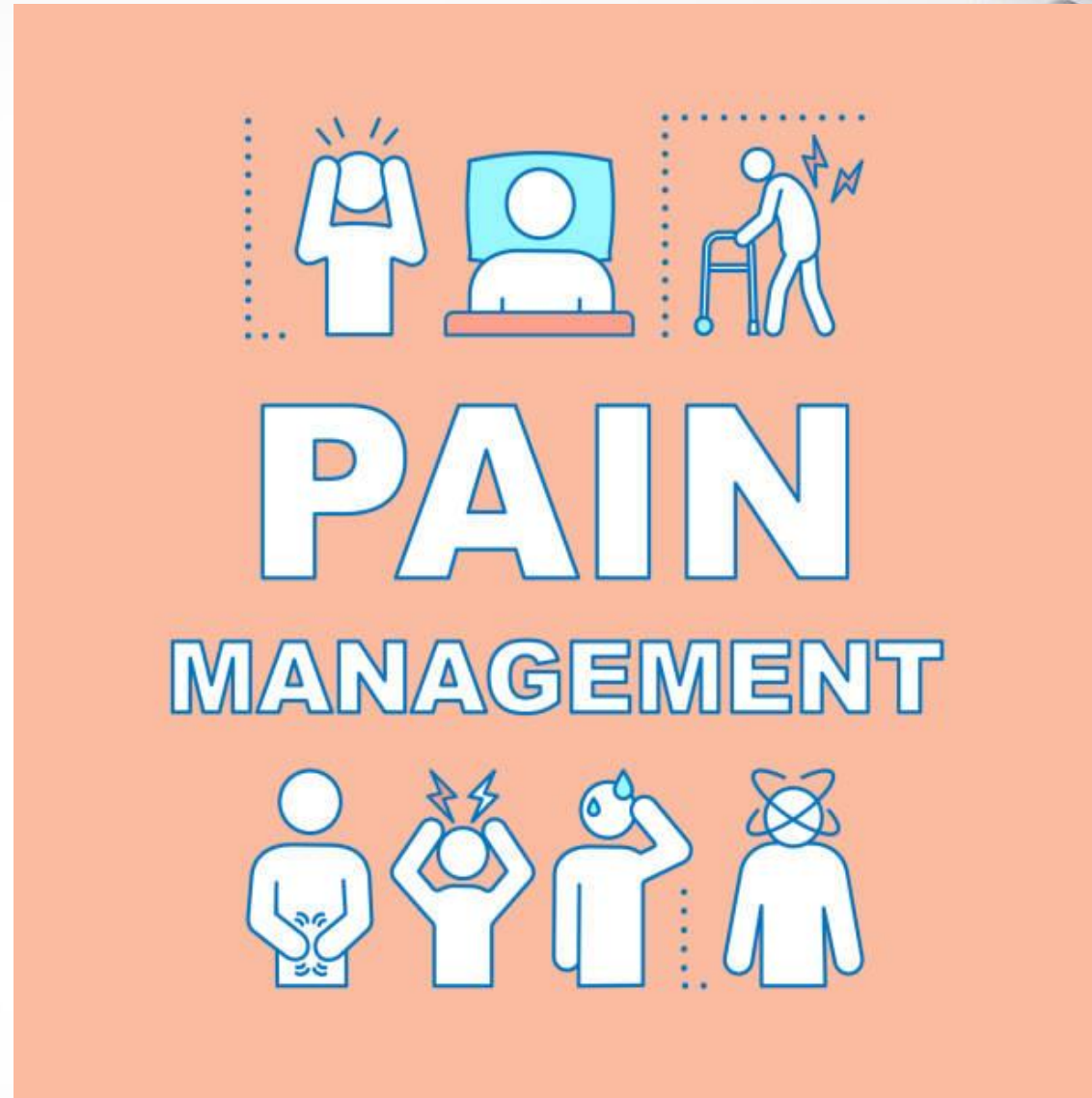


Pain Management

Prepared by
Dr : Ghada Elasyed



Intended Learning Outcomes (ILOs)



- Define Pain
- Differentiate between different types of pain
- Explain physiology of pain
- Explain tools for pain assessment
- Discuss different type of pain management

pre test



- Mention types of pain
- Mention how to assess pain for mechanical ventilated patients

Definition of Pain

- Pain is described as an unpleasant sensory and emotional experience associated with actual or potential tissue damage.



Types of Pain

Acute Pain

Chronic Pain

Nociceptive Pain

Neuropathic Pain



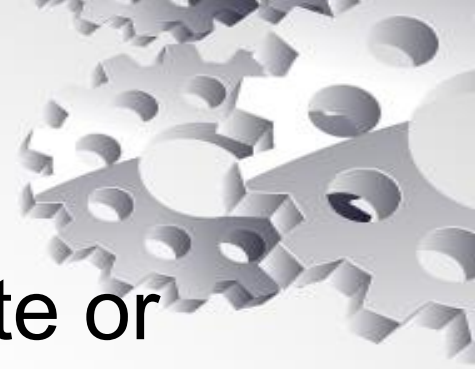
Acute Pain



- Sudden onset
- Acute pain has a short duration
- Acute pain usually corresponds to the healing process (30 days), but should not exceed 6 months.
- It implies tissue damage that is usually from an identifiable cause.
- If under treated, acute pain may bring a prolonged stress response and lead to permanent damage to the patient's nervous system.

Chronic Pain

- It develops when the healing process is incomplete or when acute pain is poorly managed
- Long term (more than 6 months), after the healing process from the original injury.
- It may or may not be associated with an illness.
- Both acute and chronic pain can have a nociceptive or neuropathic origin.



- **Nociceptive Pain**

- *Nociceptive pain arises from activation of nociceptors
- *It can be somatic or visceral.
- ***Somatic pain*** involves superficial tissues, such as the skin, muscles, joints, and bones. Its location is well defined
- ***Visceral pain*** involves organs such as the heart, stomach, and liver.

Neuropathic Pain



- **Neuropathic pain** arises from a lesion or disease affecting the somatosensory system
- The origin of neuropathic pain may be peripheral or central.
- **Neuralgia and neuropathy** are examples related to peripheral neuropathic pain treatments)



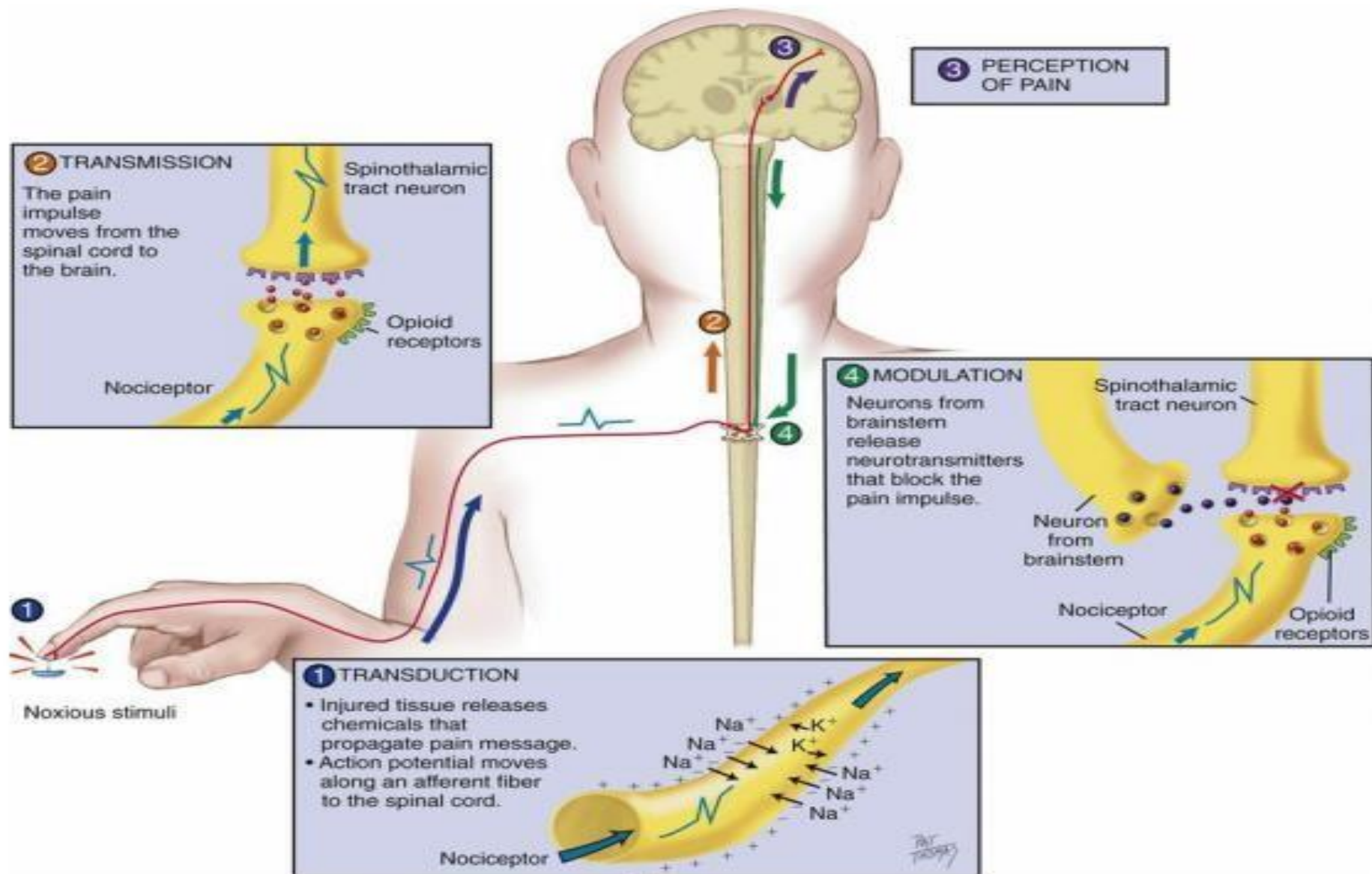
Central neuropathic pain

- involves the central somatosensory cortex and can be experienced by patients after a cerebral stroke.
- Neuropathic pain can be difficult to manage and frequently requires a multimodal approach
- (i.e., the combinations of several pharmacologic and/non pharmacologic

Physiology of Pain



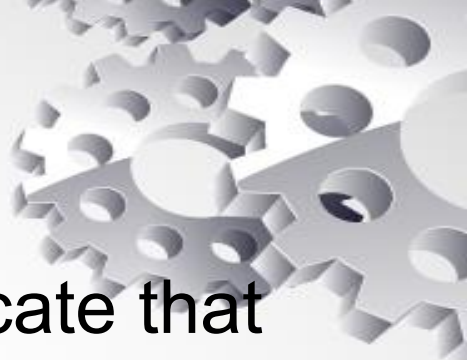
- **Four processes are involved in nociception:**
- Transduction (stimulus triggered)
- Transmission (the pain impulse moves from the spinal cord to the brain)
- Perception (represent the subjective interpretation of pain)
- Modulation (neurons from brain stem release neurotransmitters that block the pain impulse).



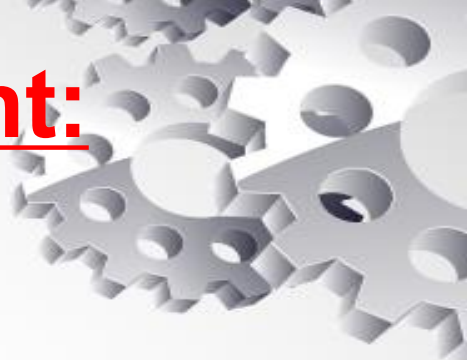
Pain Assessment



- Pain assessment is an integral part of nursing care. It is a prerequisite for adequate pain control and relief.
- Because pain is recognized as a subjective experience, the patient's self report is considered the most valid measure for pain.
- Unfortunately in critical care, many factors such as the administration of sedative agents, the use of mechanical ventilation, and altered levels of consciousness may impact communication with patients.

- 
- Many **mechanically ventilated patients** can communicate that they are in pain by using head nodding, hand motions or by seeking attention with other movements.
 - In such a situation, use of appropriate communication methods may reduce patients' distress associated with the presence of the endotracheal tube by enabling them to report the presence of pain or discomfort in a comprehensive way.

The Major Components of Pain Assessment:



- Non observable or subjective
- Observable or objective
- The Subjective Component
- Pain is known as a subjective experience.
- The subjective component of pain assessment refers to the patient's self-report about his/her sensorial, affective, and cognitive experience of pain.

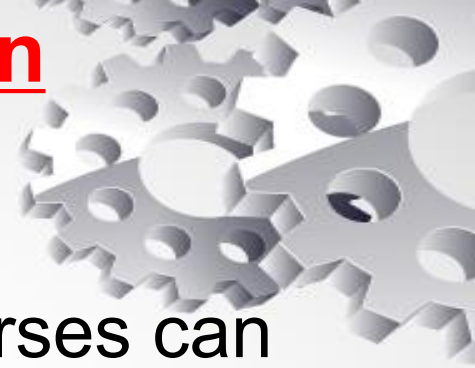
- *Mechanical ventilation should not be a barrier for nurses to document patients' self-reports of pain.
- Many mechanically ventilated patients can communicate that they have pain or can use pain scales by pointing to numbers or symbols on the scale.



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- The patient's self-report of pain can also obtained by questioning the patient using the mnemonic PQRSU

- **P**: provocative and palliative or aggravating factors
- **Q**: quality
- **R**: region or location, radiation
- **S**: severity and other symptoms
- **T**: timing
- **U**: understanding

The Observable or Objective Component of Pain Assessment



- When the patient's self-report is impossible to obtain, nurses can rely on the observation of behavioral indicators.
- Patients who experienced pain during nociceptive procedures were three times more likely to have increased behavioral responses such as facial expressions, muscle rigidity, and vocalization than patients without pain.

Tools to Tested Pain in the Critical Care Setting:



- **Behavior Pain Scale (BPS)**
- **Critical-Care Pain including Observation Tool (CPOT)**
- **Non Verbal Pain Scale (NVPS)**
- **Pain Behavioral Assessment Tool (PBAT)**
- **Pain Assessment and Intervention Notation (PAIN) Algorithm**

Behavioral Pain Scale (BPS)

- It was tested mostly in nonverbal mechanically ventilated patients with altered levels of consciousness.
- The total scores ranges from 3 to 12. A cut-off score greater than 5 indicates the presence of pain.



Behavioral Pain Scale (BPS)

Behavioral Pain Scale (BPS) Training Poster

BPS (intubated patients)

1 2 3 4

Facial expression



Relaxed Partially tightened = brow lowering Fully tightened = eyelid closing Grimacing = folded cheek

Movements of upper limbs



No movement Partially bent Very bent with finger flexion Retracted, opposition to care
At rest: check the tonus by mobilisation of the limb

Compliance with ventilation



Tolerating ventilation Coughing but tolerating ventilation most of the time Fighting ventilator but ventilation possible sometimes Unable to control ventilation

BPS-NI (non-intubated patients)

1 2 3 4

Facial expression



Relaxed Partially tightened = brow lowering Fully tightened = eyelid closing Grimacing = folded cheek

Movements of upper limbs



No movement Partially bent Very bent with finger flexion Retracted, opposition to care
At rest: check the tonus by mobilisation of the limb

Vocalisation



No pain vocalization Moaning not frequent ($\leq 3/\text{mn}$) and not prolonged ($\leq 3 \text{ s}$) Moaning frequent ($> 3/\text{mn}$) or prolonged ($> 3 \text{ s}$) Howling or verbal complaint including « Ouf! », « Ouch! » or breath-holding

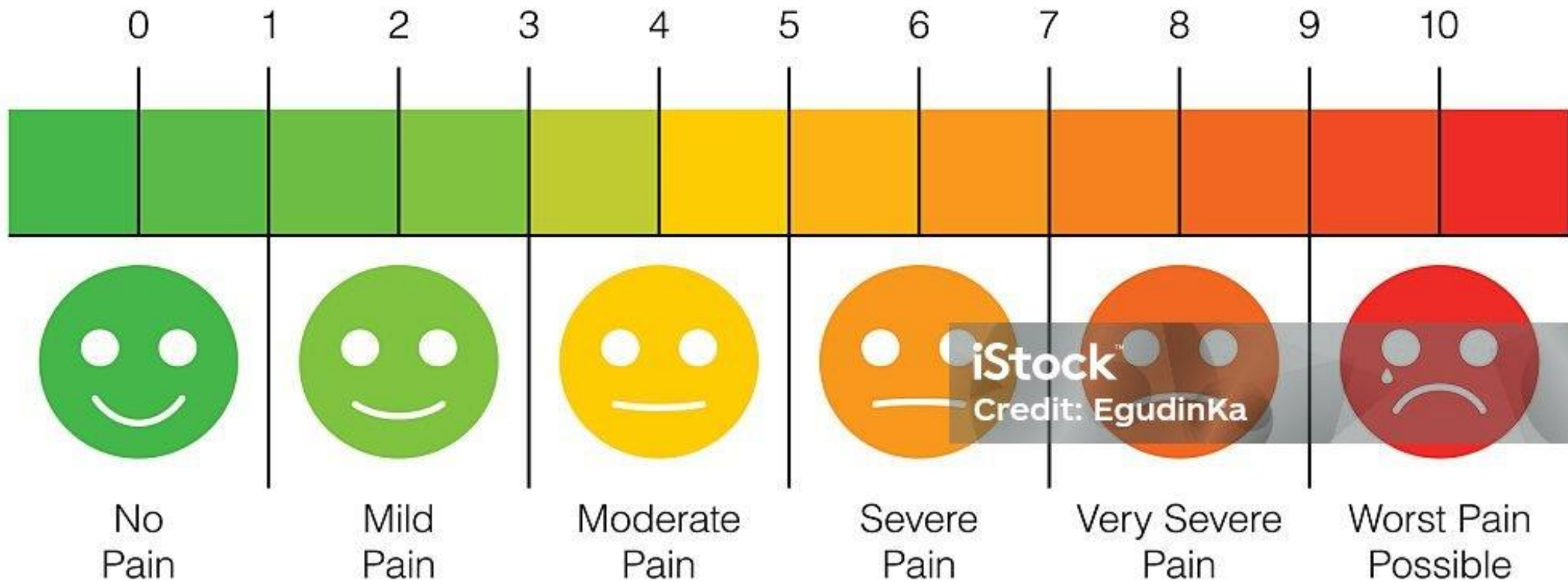
① + ② + ③ = Total BPS value

from 3 (no) to 12 (maximum) pain behavior rated using the BPS



ITEM		DESCRIPTION	SCORE
Facial Expression		*Relaxed	1
		*Partially tightened e.g., brow lowering)	2
		*Fully tightened (e.g., eyelid closing)	3
		*Grimacing	4
Upper Limbs		*No movement	1
		*Partially bent	2
		*Fully bent with finger flexion	3
		*Permanently retracted	4
Compliance Ventilation		*Tolerating movement	1
		*Coughing but tolerating ventilation for most of the time	2
		*Fighting ventilator	3
		*Unable to control ventilation	4

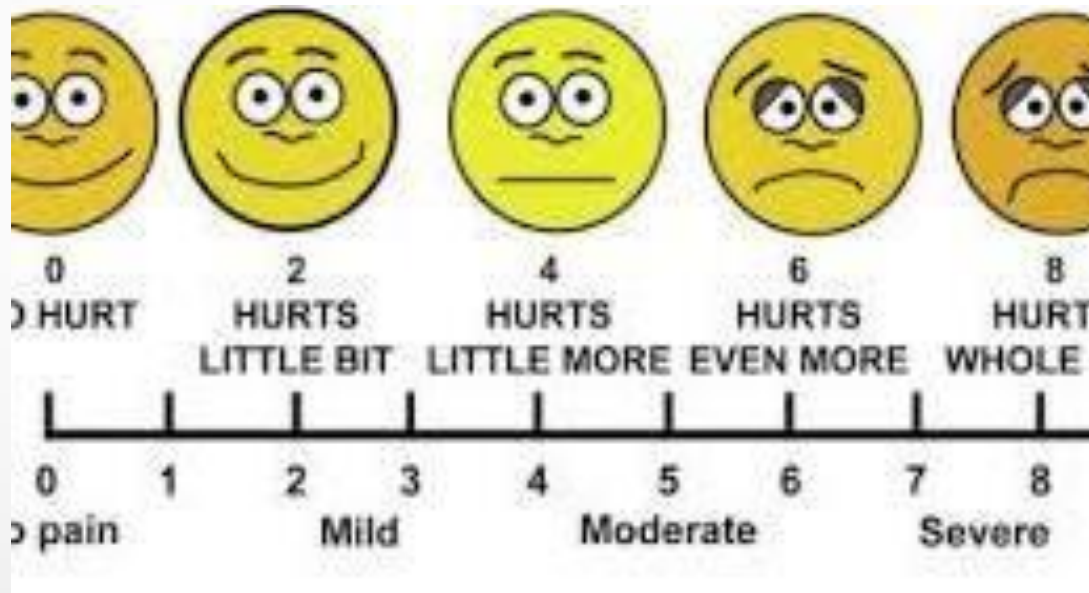
PAIN SCALE





- The **BPS** can be used quickly, it takes 2 to 5 minutes to complete, and most clinicians were satisfied with its ease of use.
- For instance,
- scores of 3 (i.e., fighting ventilator)
- 4 (i.e., unable to control ventilation) for compliance with the ventilator category may be ambiguous.
- Similarly, movements with upper limbs category may be confused with muscle tension.

Critical-Care Pain Observation Tool

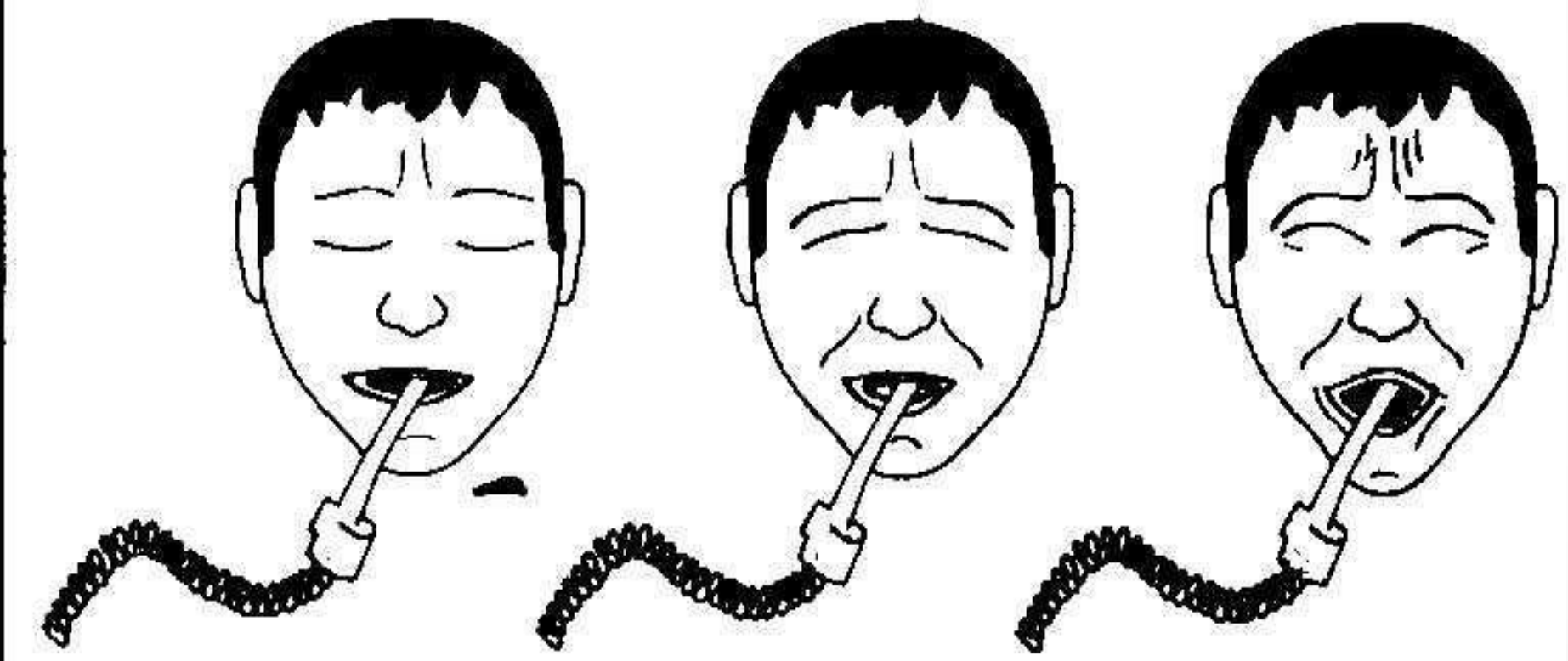


- It was tested in verbal and nonverbal, critically ill adult patients.

- **Indicator Score**

- **Facial Expression**

- Relaxed 0
- Tense 1
- Grimacing 2





Body Movements

- Absence of movements or normal position 0
- Protection 1
- Restlessness /Agitation 2

Compliance with the Ventilator (intubated patients)

- Tolerating ventilator or movement 0
- Coughing but tolerating 1
- Fighting ventilator 2



- **Vocalization (non intubated patients)**

- Talking in normal tone or no sound 0
- Sighing, moaning 1
- Crying out, sobbing 2

- **Muscle tension**

- Relaxed 0
- Tense, rigid 1
- Very tense or rigid 2

Pain Management



The control of pain can be pharmacologic, non pharmacologic pharmacological pain management

is divided into three categories of action: opioid agonists, nonopioids, and adjuvants.

1-Opioid Analgesics

The opioids most commonly used and recommended as first line analgesics are the agonists.

These opioids bind to mu receptors (transmission process) which appear to be responsible for pain relief.

Opioids should be used as first line therapy for treatment of non-neuropathic pain.

Morphine

Morphine is the most commonly prescribed opioid in the critical care unit.

Because of its water solubility, morphine has a slower onset of action and a longer duration

compared with the lipid-soluble opioids (e.g., fentanyl).

morphine-3-glucuronide (M3G, inactive) and morphine-6-glucuronide (M6G, active).

Morphine is indicated for severe pain.



- **Side Effects of Morphine**

- The hypotensive effect can be particularly problematic in the hypovolemic patient.
- The respiratory depressant effect is the most serious side effect which requiring thorough monitoring.
- Opioids may stimulate histamine release from mast cells, resulting in cardiac instability and allergic reactions.
- **Dosage**
- 1-4 mg IV bolus or 1-10 mg IV infusion.



Fentanyl

Fentanyl is a synthetic opioid preferred for critically ill patients with hemodynamic instability or morphine allergy.

It is a lipid-soluble agent that has a more rapid onset than morphine and a shorter duration.

Transdermal delivery requires 12 to 16 hours for onset of action, and it has duration of 72 hours.

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- Fentanyl is available in intravenous, intraspinal, and transdermal forms.
 - The transdermal form is commonly referred to as the Duragesic patch or the 72-hour patch.
 - When fentanyl is given by rapid administration and at higher doses, it has been associated with the **additional hazard** of bradycardia and rigidity in the chest wall muscles.
 - **Dosage**
 - 25-100 mcg IV bolus
 - 25-200 mcg IV infusion



- **Hydromorphone**
- Analgesia, anti anxiety
- **Dosage**
- 0.2 -1 mg IV bolus
- 0.2 -2 mg IV infusion
- **Side Effect of Hydromorphone:**
- Hydromorphone has the same side effects as Morphine.

- **Meperidine (Demerol)**

- It is considered the weakest of the opioids, and it must be administered in large doses to be equivalent in action to Morphine.
- A major concern with this medication, which is a Central Nervous System (CNS) neurotoxic agent.
- At high doses in patients with kidney failure or liver dysfunction or in older adult patients, it may induce **CNS toxicity**, including irritability, muscle spasticity, tremors, agitation, and seizures.



Codeine

Codeine has limited use in the management of severe pain.

It is rarely used in the critical care unit.

It provides analgesia for mild to moderate pain.

available only through oral, intramuscular, and subcutaneous routes,

Dosage: 15 - 20 mg IM, SC

Side Effects of Codeine:

the most common side effect of Codeine are lightheadedness, dizziness, shortness of breath, sedation, nausea, and vomiting.



Methadone

Methadone is a synthetic opioid with morphine- like properties but less sedation

Dosage: 5-10 mg IV. The maximum dose is 4 g/day.

Prolongation of the QT interval, which can lead to torsades de pointes, has been reported with its use.



Opioid Reversal:

Critical respiratory depression can be readily reversed with the administration of the opiate antagonist ***Naloxone***.

The usual dose is 0.4 mg, which is mixed with 10 mL of normal saline

Naloxone can be discontinued as soon as the patient is responsive to physical stimulation and able to take deep breaths.

2-Nonopioid Analgesics



Acetaminophen

It is an analgesic used to treat mild to moderate pain.

It inhibits the synthesis of neurotransmitter prostaglandins in the CNS, and this is why it has no anti-inflammatory properties.

Dosage: 650 mg. the maximum dose is 4 g/day

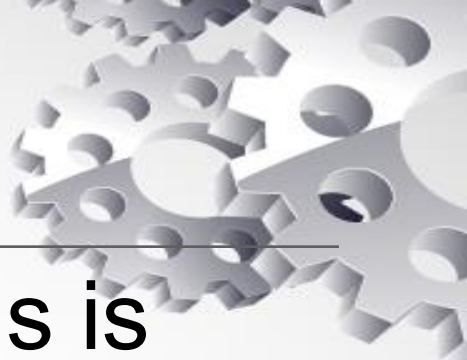
Analgesic, Antipyretic

acetaminophen total dose should not exceed 2

g/day

Side Effect: hepatotoxicity

Non Steroidal Anti-inflammatory Drugs:

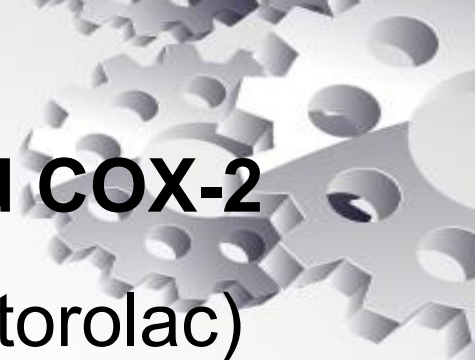


The use of NSAIDs in combination with opioids is indicated in the patient with acute musculoskeletal and soft tissue inflammation.

The mechanism of action of NSAIDs is to block the action of cyclooxygenase (COX, which has two forms:

COX-1 and COX-2), the enzyme that converts arachidonic acid to prostaglandins.

This inhibits the production of prostaglandins.

- 
- **NSAIDs** can be grouped as first-generation (**COX-I and COX-2** inhibitors, such as aspirin, ibuprofen, naproxen, and ketorolac) or second-generation (COX-2 inhibitors, such as celecoxib) agents.
 - The inhibition of COX-1 **side effects**, such as gastric ulceration, bleeding as a result of platelet inhibition and acute renal failure.
 - In contrast, the inhibition of COX-2 is responsible for the suppression of pain and inflammation.



Ketorolac is the most appropriate NSAID for use in the critical care setting.

It is a safe and effective agent for postoperative pain.

Dosage: 15-30 mg IV

Side Effect:

gastric ulceration, bleeding, exacerbation of renal insufficiency.

Use with care in older adult and renal failure patients.



- **3-Adjuvants**

- Although not widely mentioned in the critical care literature, adjuvants can be helpful for pain relief in patients with complex pain syndromes such as neuropathic pain or for other specific purposes (e.g., procedural pain).
- Anticonvulsants (e.g., carbamazepine, phenytoin, gabapentin, pregabalin) are firstline analgesics for treating neuropathic pain.

Lidocaine:

is another anesthetic that can be used for procedural and acute pain or for some patients with chronic neuropathic pain.

- When used locally, anesthetics act through the transduction process.
- The most common route for medication administration is the intravenous route by means of continuous infusion, bolus administration, or Patient-Controlled Analgesia (PCA).



Non pharmacological Methods of Pain Management

- 1-Physical Techniques
- Stimulating other non-pain sensory fibers in the periphery modifies pain transmission.
- These fibers are stimulated by thermal changes as in the application of heat or cold and simple massage.





Cold Application

Ice therapy was found to be helpful to reduce procedural pain in critically ill patients.

2-Cognitive-Behavioral Techniques

Relaxation

It is an excellent adjunct for controlling pain. Relaxation decreases oxygen consumption and muscle tone, and it can decrease heart rate and blood



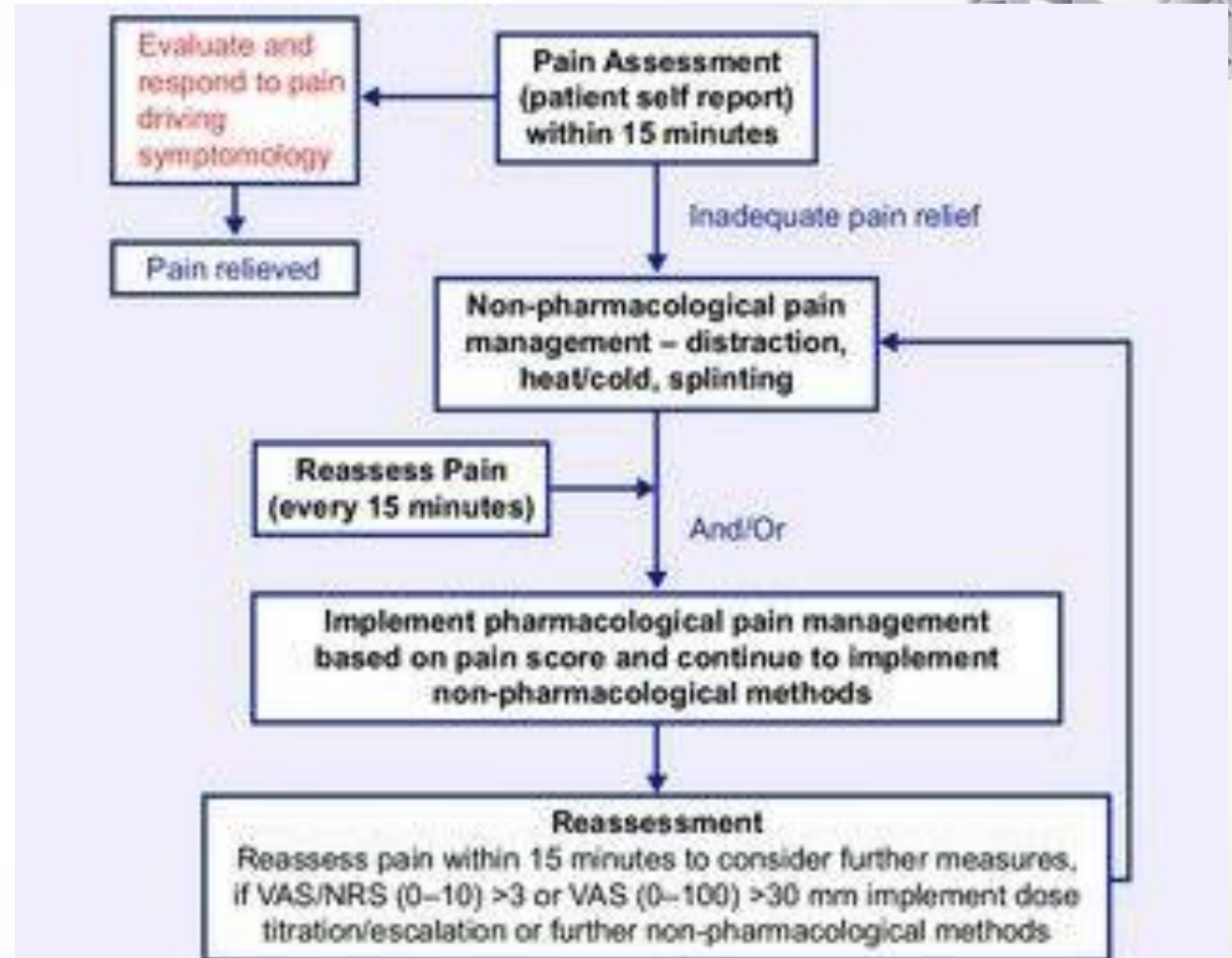
Guided Imagery

Guided imagery is a technique that uses the imagination to distract control over pain.

Music Therapy

Music therapy is a commonly used intervention for relaxation that is pleasing to the patient may have soothing effects, but its effects on reducing pain are

Role of the nurse in pain management needs to done





post test

- Mention types of pain
- Mention nonpharmacological management of pain

Thank You

